

# Practical assignment

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V1.0 (6th October)

In this course, the students must design and develop a mobile application that responds to a set of requirements.

Contrarily to traditional applications, mobile applications should be designed from the ground-up to deal, not only, with the limitations and heterogeneity of mobile devices (e.g., computation, memory, storage and power), but, above all, the characteristics and specificities of mobile devices (eg., ubiquity, touch interactions, sensor availability, etc.).

**Note:** Mobile and modern web/cloud applications have many similarities

## Application Requirements

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Design and develop a mobile application for viewing (browsing), posting (publishing) and sharing contents amongst users. The mobile application must respond to, at least, the following list of requirements:

- The application can be used by registered and unregistered users (visitors);
- All users can view and search application contents
- All users can view and search for registered users;
- Each registered user has a corresponding profile;
- User profiles must include: the user's username, a user's description, and user's published contents;
- Contents must be categorized (eg., content type);
- Contents should be presented based on content ratings, number of visualizations, number of interactions, etc. (for visitors);
- Contents can be organized based on content type;
- For registered users, contents should be displayed based on the users interests, categories, ratings or any other metric;
- Registered users can publish (add) new contents;
- Registered users can comment published contents;
- Registered users can rate published contents (eg., based on a star rating system (0 to 5 stars), likes and dislikes, etc.);
- Registered users can create interest/favorite lists (each with a custom name) and add contents to them;
- Registered users can share existing contents with other (registered) users;
- Publishing contents can be based on existing data (locally stored data) or live data (eg., live recordings);
- Published contents must be characterised by the type of content and/or other metrics;
- Contents should be sortable by publication date, rating, interactions, or other metrics;
- Records and live contents must use specific mobile device hardware during its recording (eg., GPS, accelerometer data, camera, etc.);
- The app must keep a history of a user's published contents and present statistics for those contents (in a dashboard style) (eg., number of visualizations, number of comments, histogram of ratings, etc.);
- The app must provide social style tools, such as "following" users, likes and rating contents, messaging, etc.;

- The app must use notifications for notifying users of, for example, availability of new contents;
- Application contents should be available offline; (optional)
- Application functionalities should adapt based on connectivity; (optional)
- Application fidelity should adapt based on device characteristics and specifications, connectivity, etc.; (optional)
- Application must gracefully degrade based on service availability; (optional)

## Infrastruture requirements

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The applications should be supported by an underlying distributed infrastructure. Similar to modern web applications, each functionality must be supported by an independent service (micro-services). Here are the infrastructure requirements:

- Each service must have its own design, data model and interface;
- All services should be supported by independent web services;
- All data should be accessible using REST and REST APIs (at least, one for each service);
- Each service must be supported by a set of independent Docker containers (or a single VM (not recommended));
- Services must be integrated using Docker-Compose or similar orchestration tool;
- Services must be designed and developed with gracefull degraation in mind;

## Tasks & Timeline (suggestions)

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Start by specifying what your applications will be (define the content type, target audience, etc.) **(1 week)**

Specify and design each separate functionality, as an independent services **(2 weeks)**

- Start with the infrastructure (database model, REST API, service logic)
- Only them move to the mobile components

Design and develop each functionality as an independent mobile application module **(3-4 weeks)**

- Follow a Model-View-Viewmodel (MVVM) architecture design for each module
- Use a gracefull degradation approach

Integrate the different modules into a single application **(1-2 weeks)**

## Submission and Presentation Date

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Submission and demonstration: **15th December, 2026**

Submit contents:

- Written report describing the application and including each module design, data model, APIs and interface
- Application code (link de GitHub (or similar))
- Application demonstration (video)